

## §1.4

# Data Sampling Methods

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How we choose a sample, and how it goes wrong

# Today's Goal

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- Distinguish between random samples and simple random samples
- Identify and describe different sampling methods (both random and non-random)
- Recognize sampling and non-sampling errors
- Identify different types of bias and the problems they cause

# What You Will Be Graded On

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## LT.3 Data Sampling

I understand the different sampling methods (e.g., cluster sampling, stratified sampling, etc.). I also understand the challenges associated with each of these methods.

# How Much Sleep Do Students Get?

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You want to estimate the average nightly sleep of Fresno State students.

You survey 30 students in the library at 11 p.m.

**What is wrong  
with that sample?**

The students in the library at 11 p.m. are the students who stay up late. Anyone already asleep at home had no chance of being picked.

# Two Conditions

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1. The sample should be **representative of the population**. It should reflect the key characteristics of the population, such as age distribution, gender balance, and geographic spread.
2. The sample should be **randomly selected**. Random selection helps ensure that every member of the population has a chance of being included, which reduces bias.



# Two Words, Two Meanings

## Definition 1.4.1

A **random sample** is a sample in which each member of the population has the same chance of being selected.

## Definition 1.4.2

A **simple random sample** (SRS) of size  $n$  is a sample in which every possible subset of  $n$  members has the same probability of being chosen.

# When Is a Random Sample an SRS?

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- Random sample: each *individual* has an equal chance.
- Simple random sample: each *group of that size* has an equal chance.

## Theorem 1.4.3

A random sample is also a simple random sample if the following condition is met: the selection of one member does not affect the probability of another member being selected (this is called **independence**).



# A Rule of Thumb

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In practice, if the sample is no more than 5% of the population, treat a random sample as an SRS.

Fresno State has about 24,000 students. A random sample of 200 is under 1%, so treat it as an SRS.



# Sampling Whoever Is There

## Definition 1.4.4

**Convenience sampling** is a method of sampling in which the sample is chosen based on ease of access or availability.

## Example 1.

A student wants to know how Fresno State students feel about campus parking, so she surveys the other 34 students in her own 8 a.m. lecture.

# Why That One Is Risky

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## Warning

Convenience samples are generally not reliable for making inferences about a population. They should be avoided when accurate conclusions are needed.

**Problem:** Convenience samples often lead to **selection bias** because the easily accessible group may differ systematically from the rest of the population.

# Pick a Start, Then Count Off

## Definition 1.4.7

**Systematic sampling** is a method of sampling in which we select a random starting point and then select every  $k$ th member of the population.



Start at 2, then every 3rd member: 2, 5, 8, 11, ...

# Systematic in Practice

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## Example 2.

At a campus health fair, a nurse picks a random person in line and then screens every 3rd person after that.

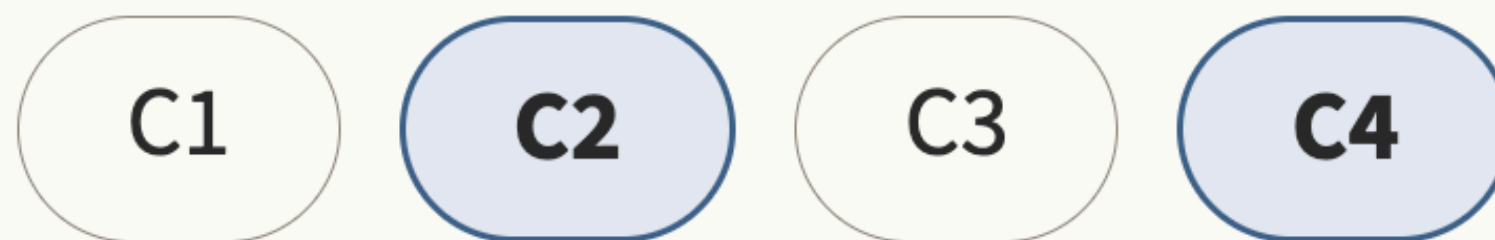
Is this a simple random sample?

**No.** Once the start is chosen, the rest is fixed. Starting at 2 and taking every 3rd, no sample can hold both person 2 and person 3.

# Pick Some Groups, Take Everyone

## Definition 1.4.10

**Cluster sampling** is a method in which the population is divided into groups called **clusters**. Some clusters are randomly selected, and then *all* members from the selected clusters are included in the sample.



Sample = **all** members of C2 and C4



# Cluster in Practice

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## Example 3.

To study 4th-grade reading, a district randomly selects 6 of its 42 elementary schools and then tests *every* 4th grader at those 6 schools. The schools are the clusters.

Is this a simple random sample?

**No.** Members of one cluster always travel together. If A and B attend the same school, no sample holds A but not B.



# Pick From Every Group

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## Definition 1.4.13

**Stratified sampling** is a method in which the population is divided into subgroups called **strata** based on shared characteristics (e.g., age, gender, income level). Then a random sample is selected from *each* stratum.



# Four Strata, One Sample

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Stratum 1 — first-year students

**take 25 at random**

Stratum 2 — sophomores

**take 25 at random**

Stratum 3 — juniors

**take 25 at random**

Stratum 4 — seniors

**take 25 at random**

Sample = the four selections combined

# Stratified in Practice

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## Example 4.

The advising center wants to know how well students understand their degree requirements. Understanding changes a lot from year to year, so they split students by class standing and randomly select 25 from *each* group.

Is this a simple random sample?

**No.** Every stratum is guaranteed a seat. No sample can come entirely from one class standing.

# The Two That Get Mixed Up

| Cluster Sampling                                        | Stratified Sampling                             |
|---------------------------------------------------------|-------------------------------------------------|
| Randomly select <b>some</b> clusters                    | Select from <b>each</b> stratum                 |
| Include <b>ALL</b> members of selected clusters         | Include <b>SOME</b> members from each stratum   |
| Clusters should be representative of population         | Strata are defined by shared characteristics    |
| More practical when population is geographically spread | Better for ensuring representation of subgroups |

# Now You Try: Name the Method

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**You Try 1.4.1.** Name the sampling method used in each scenario.

1. A pollster stands outside one grocery store and interviews shoppers as they leave. **Convenience.**
2. A factory tests every 50th item off the line, starting from a randomly chosen item. **Systematic.**
3. A city picks 8 of its 90 neighborhoods at random and surveys every household in those 8. **Cluster.**
4. A hospital divides nurses by shift and randomly picks 10 from each shift. **Stratified.**

# The Error You Cannot Avoid

## Definition 1.4.17

**Sampling error** is the natural variation between a sample statistic and the population parameter, caused simply by the fact that we are using a sample instead of the entire population.

**Problem caused:** Sampling error is unavoidable when using samples, but it can be **reduced by using larger sample sizes**. It is not a “mistake” — it is a natural consequence of sampling.



# The Error You Can Avoid

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## Definition 1.4.18

**Non-sampling error** refers to errors that are not related to the process of sampling, such as bias in the survey design, data entry mistakes, or problems with how questions are asked.

These come from **bias**. The rest of today names the six kinds.



# Where Should the Armor Go?

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In World War II, analysts examined bombers that came back. The bullet holes clustered on the wings and the tail.

**Where would you  
put the armor?**

Abraham Wald said: armor the engines and the cockpit — the places with *no* holes.

Planes hit there never came back. The sample was every plane that *survived*.

# Who Got In, and Who Answered

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- 1. Selection Bias:** Occurs when the sample is not representative of the population, leading to over- or under-representation of certain groups.  
*Problem:* Results cannot be generalized to the entire population. Conclusions may only apply to the overrepresented group.
- 2. Non-response Bias:** Occurs when individuals selected for the sample fail to respond, and those who don't respond differ systematically from those who do.  
*Problem:* The sample becomes unrepresentative because certain perspectives are missing. For example, busy people or those with negative opinions may be less likely to respond.

# What They Said, and Who Spoke Up

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- 3. Response Bias:** Occurs when respondents provide inaccurate information, intentionally or unintentionally.

*Problem:* Data is systematically distorted. Common causes include social desirability (giving “acceptable” answers), poor memory, or misunderstanding questions.

- 4. Voluntary Response Bias:** Occurs when a sample consists of people who choose to participate (e.g., online polls, call-in surveys).

*Problem:* People with strong opinions (often negative) are more likely to volunteer, leading to extreme results that don’t reflect the general population.

# Who Survived, and How You Asked

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**5. Survivorship Bias:** Occurs when data only includes subjects that “survived” a selection process, ignoring those that didn’t.

*Problem:* Leads to overly optimistic or skewed conclusions. For example, studying only successful companies ignores lessons from failed ones.

**6. Leading Question Bias:** Occurs when question wording influences respondents toward a particular answer.

*Problem:* Responses reflect the question’s suggestion rather than the respondent’s true opinion, invalidating the results.



# Name the Bias, Parts a and b

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- a. A survey asks employees how many workdays they called in sick without being ill. **Response bias.** *Problem:* the count is systematically understated, so the results are unreliable.
- b. A gym reports that 88% of members improved their fitness, based on a survey of people who currently hold memberships. **Survivorship bias.** *Problem:* the success rate is inflated, because members who quit are never surveyed.

# Name the Bias, Part c

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- c. A survey asks: “Do you support the wasteful new stadium that will raise your rent?” **Leading question bias.** “Wasteful” and “raise your rent” push toward one answer.

*Problem:* the responses show artificially high opposition, not real opinion.